# Applied Machine Learning

# MLR3 Pipelines: Part 1

**Learning goals**

- Overview of PipeOps
- Nonlinear Pipelines
- Pipeline Construction

# “PIPELINES” DICTIONARY & SHORT FORM

### Many frequently used *patterns* for pipelines

- Making Learners robust to bad data (imputation + feature encoding + . . . )
- Bagging
- Branching

### Collection of these is in mlr3pipelines

```
head(as.data.table(mlr_pipeops), 5)[, list(key, input.num, output.num)]

#> Key: <key>
#>               key input.num output.num
#>            <char>     <int>      <int>
#> 1:         boxcox         1          1
#> 2:         branch         1         NA
#> 3:          chunk         1         NA
#> 4: classbalancing         1          1
#> 5:     classifavg        NA          1
```

# “PIPELINES” DICTIONARY & SHORT FORM

### Many frequently used *patterns* for pipelines

- Making Learners robust to bad data (imputation + feature encoding + ...)
- Bagging
- Branching

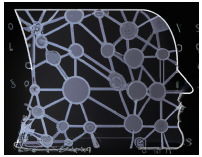
### Collection of these is in mlr3pipelines

`po()` accesses the `mlr_pipeops` “Dictionary”.

```
pca = po("pca")
pca

#> PipeOp: <pca> (not trained)
#> values: <list()>
#> Input channels <name [train type, predict type]>:
#>   input [Task,Task]
#> Output channels <name [train type, predict type]>:
#>   output [Task,Task]
```

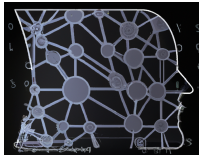
# PIPEOPS SO FAR



```
mlr_pipeops$keys()
```

```
#>  [1] "boxcox"                  "branch"                  "chunk"
#>  [4] "classbalancing"          "classifavg"              "classweights"
#>  [7] "colapply"                "collapsefactors"         "colroles"
#> [10] "copy"                    "datefeatures"            "encode"
#> [13] "encodeimpact"            "encodelmer"              "featureunion"
#> [16] "filter"                  "fixfactors"              "histbin"
#> [19] "ica"                     "imputeconstant"          "imputehist"
#> [22] "imputelearner"           "imputemean"              "imputemedian"
#> [25] "imputemode"              "imputeoor"               "imputesample"
#> [28] "kernelpca"               "learner"                 "learner_cv"
#> [31] "missind"                 "modelmatrix"             "multiplicityexply"
#> [34] "multiplicityimply"       "mutate"                  "nmf"
#> [37] "nop"                     "ovrsplit"                "ovrunite"
#> [40] "pca"                     "proxy"                   "quantilebin"
#> [43] "randomprojection"        "randomresponse"          "regravg"
#> [46] "removeconstants"         "renamecolumns"           "replicate"
#> [49] "scale"                   "scalemaxabs"             "scalerange"
#> [52] "select"                  "smote"                   "spatialsign"
#> [55] "subsample"               "targetinvert"            "targetmutate"
#> [58] "targettrafoscalerange"   "textvectorizer"          "threshold"
#> [...]
```

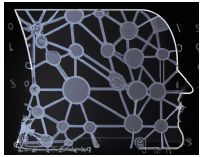
# PIPEOPS SO FAR AND PLANNED

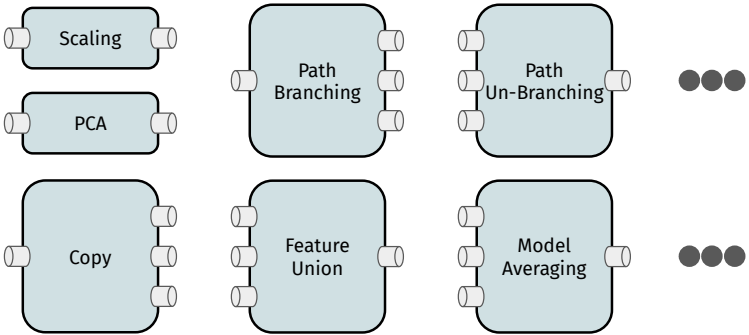


- Simple data preprocessing operations (scaling, Box Cox, Yeo Johnson, PCA, ICA)
- Missing value imputation (sampling, mean, median, mode, new level, ...)
- Feature selection (by name, by type, using filter methods)
- Categorical data encoding (one-hot, treatment, impact)
- Sampling (subsampling for speed, sampling for class balance)
- Ensemble methods on Predictions (weighted average, possibly learned weights)
- Branching (simultaneous branching, alternative branching)
- Combination of data: `featureunion`
- Text processing
- Date processing
- Time series and spatio-temporal data *(planned)*
- Multi-output and ordinal targets *(planned)*
- Outlier detection *(planned)*

# Nonlinear Pipelines

# PIPEOPS WITH MULTIPLE INPUTS / OUTPUTS

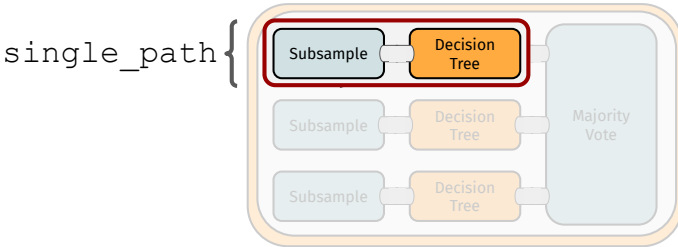




# MLR3PIPELINES IN ACTION

## Ensemble Method: Bagging

```
single_path = po("subsample") %>>% lrn("classif.rpart")
```

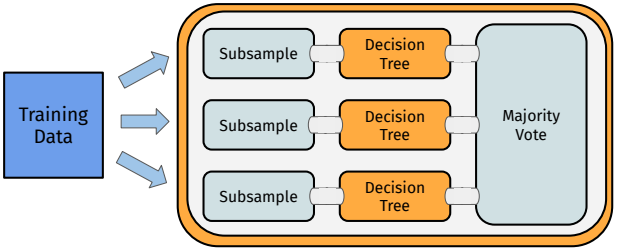


# MLR3PIPELINES IN ACTION

## Ensemble Method: Bagging

```
single_path = po("subsample") %>>% lrn("classif.rpart")

graph_bag = pipeline_greplicate(single_path, n = 3) %>>%
  po("classifavg")
```



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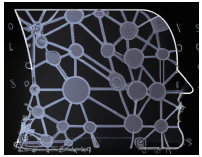

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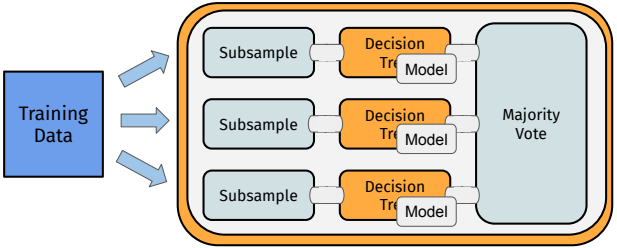
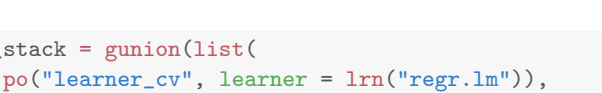


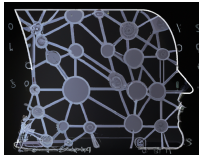
# MLR3PIPELINES IN ACTION

## Ensemble Method: Stacking

```
graph_stack = gunion(list(
      po("learner_cv", learner = lrn("regr.lm")),
      po("learner_cv", learner = lrn("regr.svm")),
      po("nop"))) %>>%
  po("featureunion") %>>%
  lrn("regr.ranger")
```



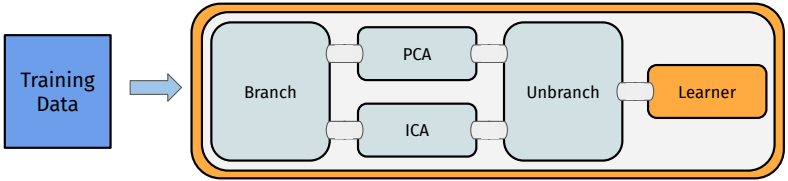
# MLR3PIPELINES IN ACTION



## Branching

```
graph_branch = ppl("branch", list(
  pca = po("pca"),
  ica = po("ica"))) %>>%
  lrn("classif.kknn")
```

Execute only one of several alternative paths



# MLR3PIPELINES IN ACTION

## Branching

```
graph_branch = ppl("branch", list(
  pca = po("pca"),
  ica = po("ica"))) %>>%
  lrn("classif.kknn")
```

```
> graph_branch$pipeops$branch$
    param_set$values$selection = "pca"
```

# MLR3PIPELINES IN ACTION

## Branching

```
graph_branch = ppl("branch", list(
  pca = po("pca"),
  ica = po("ica"))) %>>%
  lrn("classif.kknn")
```

```
> graph_branch$pipeops$branch$
    param_set$values$selection = "ica"
```